

Student 2 - Java Concepts & Class Design

Slides 4–5 | Time: 2 min 30 sec | Role: Technical depth, show OOP knowledge

"So now that you know what the system does — let me show you how it is built."

Click to Slide 03 — Java Concepts Used. Stand to the left of the screen.

"We used nine Java concepts in this project. Each card shows the actual syntax we wrote."

Sweep hand across the grid of 9 cards.

"The first one is the basics — variables and data types. We use int for IDs, Strings for names, booleans for status."

Point at Card 1 — STRING badge. Point at the code snippet.

"Then we need to store multiple items. That is where ArrayList comes in — List of Item, created with new ArrayList."

Point at Card 2 — LIST badge.

"We use static methods in our Validator class. You do not need to create an object — just call Validator dot isValidDate directly."

Point at Card 3 — STATIC badge.

"Conditional statements — if the user types 1, we post a lost item. If they type 8, we exit."

Point at Card 4 — IF/ELSE badge.

"This is the big one — classes and objects. Our project has six classes. Each one has a single job."

Point at Card 5 — 6 CLASSES badge. Emphasize "six" with your fingers.

"Encapsulation — we keep fields private and provide public getters. Only status has a setter because that is the only thing that changes."

Point at Card 6 — PRIVATE badge.

"Loops — while true keeps the menu running. For-each loops let us go through every item in the list."

Point at Card 7 — WHILE/FOR badge.

"Constructors — when we create a new Item, the constructor sets all eight fields at once using the this keyword."

Point at Card 8 — ITEM badge.

"And finally — streams and lambdas. Instead of writing long loops to filter items, we use items dot stream dot filter dot collect. Clean, short, readable."

| Point at Card 9 — STREAM badge. Make a flowing motion with your hand to show the pipeline.

"Now let me show you how these classes fit together."

| Click to Slide 04 — Class Design. The UML diagram appears.

Smooth transition from concepts to architecture.

"This is our UML class diagram. Six classes — and the data file."

| Point at the entire diagram with a sweep gesture.

"Main sits at the top. It handles the menu and user input. It has a Scanner and an ItemService."

| Point at the Main class box.

"Main never touches the file directly. It calls ItemService."

| Trace the arrow from Main to ItemService. Shake head slightly on "never touches the file."

"ItemService is the brain. It has postItem, getLostItems, getFoundItems, and claimItem."

| Point at the ItemService class box.

"ItemService never reads user input. It only processes data."

| Point at ItemService methods. Nod for emphasis.

"StorageManager handles all file operations — load, append, and rewrite. It reads and writes to data slash items dot txt."

| Point at StorageManager class box, then at the data node.

"Item is the data model — eight fields, getters, one setter, and two formatting methods."

| Point at the Item class box showing fields and methods.

"Validator is a standalone utility. It depends on nothing. Just two static methods for checking input."

| Point at Validator class box.

"And Tests is our test harness — passed and failed counters, and a check method."

| Point at Tests class box.

"The key rule — each class has one job. Main does UI. ItemService does logic. StorageManager does files. That is separation of concerns."

| Point at each layer as you name it. Make a chopping motion for "one job."

"That covers the design. Now over to explain the actual implementation code."

| Turn towards next speaker. Step back.